

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Sanjanaa Sree .P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Unit 1 : Assessment Tool 2

Constraint - Based Problem Solving

Course Code : CBA1710 Artificial Intelligence

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1). Map Coloring using Backtracking Search (CSP) :

Given Regions :

Regions : R_1, R_2, R_3, R_4

Colors : {Red, Green, Blue}

Constraints :

(*) Adjacent regions cannot have the same color.

$$R_1 \leftrightarrow R_2$$

$$R_2 \leftrightarrow R_3$$

$$R_3 \leftrightarrow R_4$$

$$R_4 \leftrightarrow R_1$$

(*) $R_1 = \text{Red}$.

(*) $R_3 \neq \text{Blue}$.

(*) Every region gets exactly one color.

Step 1 : Assign R_1

Given constraints :

$$R_1 = \text{Red}$$

Region	Color
R_1	Red
R_2	?
R_3	?
R_4	?

Constraint Check : Valid .

Step 2 : Assign R_2

Possible colors for $R_2 = \{\text{Green, Blue}\}$

$$R_2 = \text{Green}$$

Region	Color
R_1	Red
R_2	Green
R_3	?
R_4	?

Constraint Check : $R_1 \neq R_2$

Step 3: Assign R3

Constraint: R3 cannot be blue.

Possible colors = {Red, Green}

R3 = Red

Region	Color
R1	Red
R2	Green
R3	Red
R4	?

Constraint check: $R2 \neq R3$ (Green $\neq$ Red)

R3 is not blue.

Step 4: Assign R4

Adjacent to R3 and R1.

Possible colors = ~~{Green, Blue}~~

R4 = Green

Region	Color
R1	Red
R2	Green

R_3

Red

Green

R_4

Constraint check : $R_3 \neq R_4$

$R_4 \neq R_1$

All constraints are satisfied.

Final Answer :

$R_1 = \text{Red}$, $R_2 = \text{Green}$, $R_3 = \text{Red}$, $R_4 = \text{Green}$.

2). Delivery Robot using BFS Search :

S	.	.	X	.
.	X	.	X	.
.	X	.	.	.
.	.	X	X	.
.	.	.	.	G

where : $S = \text{Start } (0,0)$

$G = \text{Goal } (4,4)$

$X = \text{Obstacle}$

Manhattan Distance formula :

$$h(n) = |x_g - x_n| + |y_g - y_n|$$

Goal = (4,4)

SFS Search Steps :

Initial State :

Frontier = $[(0,0)]$

Visited = $\{(0,0)\}$

Expand $(0,0)$

Possible moves :

(*) Right $\rightarrow (0,1)$

(*) Down $\rightarrow (1,0)$

Frontier : $[(0,1), (1,0)]$

Expand $(0,1)$

Valid :

Right $\rightarrow (0,2)$

Frontier : $[(1,0), (0,2)]$

Expand $(1,0)$

Valid :

Down $\rightarrow (2,0)$

Frontier : $[(0,2), (2,0)]$

Expand $(0,2)$

Valid :

Down $\rightarrow (1,2)$

Frontier : $[(2,0), (1,2)]$

Expand $(2,0)$:

Valid :

Down $\rightarrow (3,0)$

Frontier : $[(1,2), (3,0)]$

Expand $(1,2)$:

Valid :

Down $\rightarrow (2,2)$

Frontier : $[(3,0), (2,2)]$

Expand $(3,0)$:

Valid :

Right $\rightarrow (3,1)$

Down $\rightarrow (4,0)$

Frontier :

~~$[(2,2), (3,1), (4,0)]$~~

Expand $(2,2)$:

Valid :

Right $\rightarrow (2,3)$

$[(3,1), (4,0), (2,3)]$

Expand (3,1) :

Valid :

Down $\rightarrow$ (4,1)

Frontier :

$[(4,0), (2,3), (4,1)]$

Optimal Path :

(0,0)

$\rightarrow$ (0,1)

$\rightarrow$ (0,2)

$\rightarrow$ ~~(1,2)~~ (1,2)

$\rightarrow$ (2,2)

$\rightarrow$ (2,3)

$\rightarrow$ (2,4)

$\rightarrow$ (3,4)

$\rightarrow$ (4,4)

Expand (2,3) :

Valid :

Right $\rightarrow$ (2,4)

Frontier :

$[(4,1), (2,4)]$

Cost Calculation :

Expand (2,4) :

No. of moves = 8

Valid :

Cost per move = 1

Down $\rightarrow$ (3,4)

Total Cost = $8 \times 1 = 8$

Frontier :

$[(4,1), (3,4)]$

Expand (3,4) :

Valid :

Down $\rightarrow$ (4,4)

Frontier :

$[(4,1), (4,4)]$

$\Rightarrow$ Goal found.